IAEA Coordinated Research Project on Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments

Test Procedure for Semiconductor Charged Particle Radiation Detectors

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1.- OBJECTIVE AND SCOPE

1.1.- Objective

The objective of this procedure is to describe the steps to verify the radiation response and the electrical characteristics of systems with charged particle semiconductor detectors employed in counting and spectroscopy of charged particles.

1.2.- Scope

This procedure is applicable only for the verification and diagnostic of charged particle detectors such as the silicon surface barrier detectors and ion–implanted detectors.

2.- NOTATION AND DEFINITIONS

2.1.- Notation

SSB	Silicon Surface barrier detectors
MCA	Multichannel Analyzer
DMM	Digital multimeter

2.2.- Definitions

2.2.1. Detector
A device that converts the energy of a photon or incident particle in an electric pulse.

2.2.2. Silicon Surface Barrier detector
A radiation detector that is made with high resistivity silicon and a very thin rectifying contact made by evaporation or diffusion.

2.2.3 Ion-implanted detector
A radiation detector that is made with high resistivity silicon and a very thin rectifying contact made by ion implantation.

2.2.4. Dead Layer
A region in the detectors in which no useful ionization is produced.

2.2.5 Detector Background Counting
Counting that is present even without a radiation source, when the detector is biased.

2.2.6. Resolution
Capability of a radiation detector to distinguish between two peaks of next energies. For charged particle detectors, it is expressed in keV´s.

2.2.7. Charged particles
The charged particles considered in this procedure are: electrons, protons, alpha and beta particles and nuclei of light and heavy elements.

3.- DEVELOPMENT.

The sequence of steps to verify the radiation response and electrical characteristics of charged particle semiconductor detectors is described in the next paragraphs. A flux diagram of the process is shown in Annex I.

3.1- Generalities

A technical report of the verification of the electrical characteristics and the response to radiation of charged particle semiconductor detectors must include the circuit diagram, environmental conditions, geometry of the testing set-up, count rate, isotope and test instruments employed.

It is assumed that the detector has a bias resistor with a value according to the leakage current in order to generate a voltage drop less than 10% of the applied voltage.

3.2- Introduction

Semiconductor charged particle spectrometers consist of a semiconductor radiation detector, signal processing electronics interfaced to a pulse height analyzer and a computer, see Fig.1.

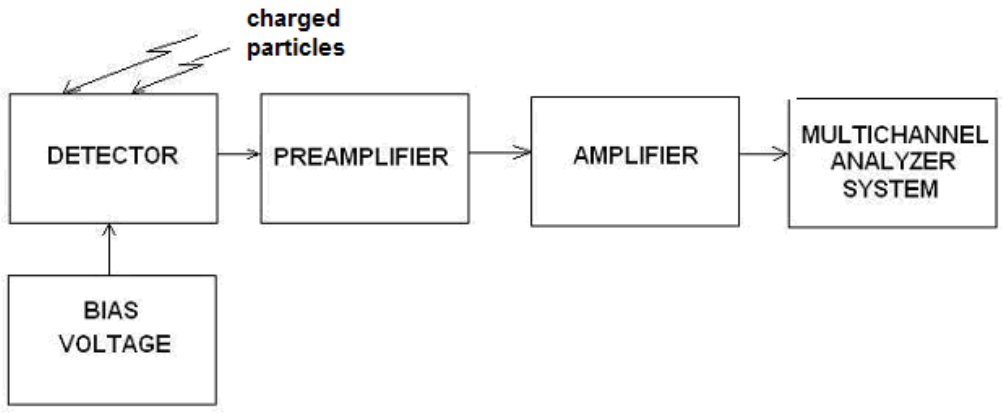


Fig. 1.- Blocks diagram of a Charged particle spectrometer

The detector is a semiconductor crystal between two conductor electrodes. A potential difference is established between the electrodes thereby producing an electric field in the semiconductor. A charged particle has a very great ionisation capability, when enters the semiconductor, it interacts and produces free charge carriers in the crystal, the number of which is proportional to the energy lost by the particle. The charge motion resulting from the influence of the electric field produces an induced current pulse in the external circuit. The integrated current pulse is proportional to the energy lost by the charged particles.

The pulses are routed to a multichannel pulse-height analyzer (MCA) where they are sorted and stored according to the amplitude distribution to produce a pulse-height graph that corresponds to the energy spectrum of the charged particles. The MCA may

be a dedicated instrument or an analog-to-digital converter (ADC) interfaced to a computer.

The semiconductor material most frequently used for charged particles spectrometers, is silicon with high resistivity to withstand large electric fields without excessive leakage currents. In order to limit the current flow through the semiconductor device, a reverse biased diode junction is utilized.

The noise in the detector and in the preamplifier circuit is not significant in this case, if it is compared with the size of the signal produced, because the measured particles generally have high energies. Consequently, both the semiconductor detector and the preamplifier are usually operated at room temperature.

Energy resolution is one of the most important characteristics of a charged particle energy spectrometer since it sets the limit on the ability to resolve closely spaced lines. Other important parameters are count-rate capability, gain stability, and detector size (efficiency).

Effects of high count rate such as pulse pile-up and dc level shifts need to be considered. Every time an event produces a pulse in the amplifying equipment, the dc levels throughout the system are perturbed and take some time to return to their original values. If another event occurs within this time interval, its effective output pulse height may be altered, thereby contributing to spectral distortion.

3.3.- Test Instruments

All the instruments employed in the tests must be calibrated and with a valid calibration certificate.

3.3.1 Detector Bias Power Supply.

Generally it is a high voltage power supply with low current capability to feed the detector, regulation less than 0.1 % is desired. The ripple should be less than 100 mV.

3.3.2 Pulse Amplifier.

A spectroscopy amplifier with shaping times less than 3 μs is required.

3.3.3 Oscilloscope

Use an analog or digital oscilloscope.

3.3.4 Counter or Scaler

The number of counts per time unit is measured with a counter (scaler) or rate meter for nuclear pulses if a counting system is assembled, it generally includes a voltage discriminator to block low amplitude noise pulses. The counter measures positive pulses.

3.3.5 Preamplifier

A charge sensitive preamplifier is the best option, generally the bias voltage pass through a filter that is inside the preamplifier, also a test input is available. Examples of available preamplifiers are: ORTEC 142, ORTEC 109A, CANBERRA 2003BT, CANBERRA 2004.

3.3.6. Multichannel Analizer, MCA

A MCA with at least 1024 channels is needed to get the energy spectrum. Consider that the analog input pulse must be positive.

3.4.- Test Conditions

3.4.1.- Background Radiation

Be sure that the only contribution to the detector counting is the natural background, avoiding the contribution due to any additional radioactive source.

3.4.2.- Temperature

Some characteristics of charged particle semiconductor detectors are temperature dependant, a reference temperature between 20º C and 25° C is recommended.

3.4.3.- Light leakage

Charged particle semiconductor detectors are very sensitive to the visible light, then any light leakage must be avoided.

3.4.4.- Vacuum conditions

In the measurement of charged particles, the air interacts with the particles, reducing its apparent energy, in order to get the best results, a good vacuum, better than 1×10^{-2} Pa (1×10^{-4} mbar), must be established in the measuring chamber.

3.4.5.- Radiation flux

Charged particle semiconductor detectors can not operate at very high radiation fluxes because they has a limitation related with the number of incident particles per unit of time and its amount of energy, in normal applications the radiation flux is less than 100 000 MeV/s.

3.4.6.- Radioactive source employed

Generally, alpha radiation from an Am-241 radioactive source is specified to characterize the charged particle detectors.

The energy calibration of the spectroscopy system can be done with a triple source containing Am-241, Pu-239 and Cm-244. See Table 1, for details. Sources with an activity of around 37 000 Bq (1 μCi) could be used.

Isotope	Radiation	Energy (MeV) (% yield)	Mean life
^{239}Pu (Plutonium)	alpha	5.104 (11.5 %) 5.142 (15.1 %) **5.155 (73.3 %)**	24 000 years
^{241}Am (Americium)	alpha	5.389 (1.3 %) 5.443 (12.7 %) **5.486 (86 %)**	458 years
^{244} Cm (Curium)	alpha	5.762 (23.6 %) **5.804 (76.4 %)**	18 years

Table 1. Radioactive sources used for energy calibration.

3.5.- Test Circuits

Whenever possible, refer to the test conditions recommended by the detector's manufacturer in the specifications sheet, if it is not available, use the diagram shown in Fig. 1.

3.6.- Measurements

In this paragraph, the measuring techniques employed to obtain the parameters of charged particle semiconductor detectors are defined.

3.6.1 Measurement of the rectifying junction.

The rectifying junction of the detector can be measured with a DMM, in the diode option, in forward bias condition, the reading should be around 0.6 V. For thicker detectors, this value should be larger and the rectifying junction could be measured with a curve tracer, see Fig. 2. In reverse bias condition, the behavior is almost like an open circuit, because the leakage current is very small (around 10 nA or less).

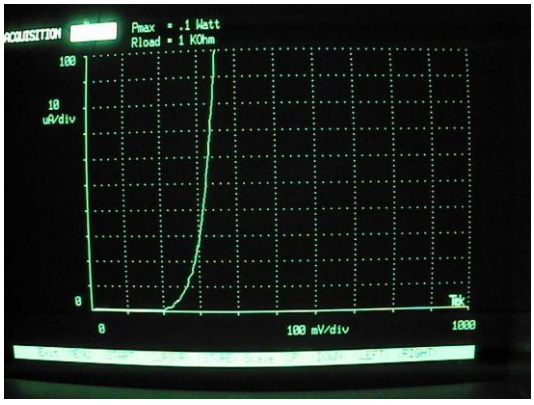


Fig. 2.- I-V curve for a SSB, the ranges in the curve tracer are: 10 μA/div., 100 mV/div.

3.6.2. Reverse current.

The leakage or reverse current in a charged particle semiconductor detector must be less than 100 nA, it depend on the characteristics of the detector and its area. Fig. 3 shows the circuit used to make the measurement of reverse current. An electrometer is required (for example: Keithley 610C, Keithley 6517A) and a voltage power supply. If light is entering the detector surface, it contributes drastically to the increase in the reverse current.

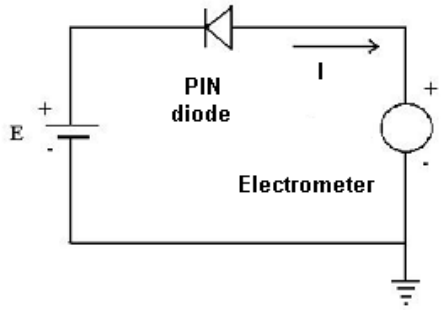
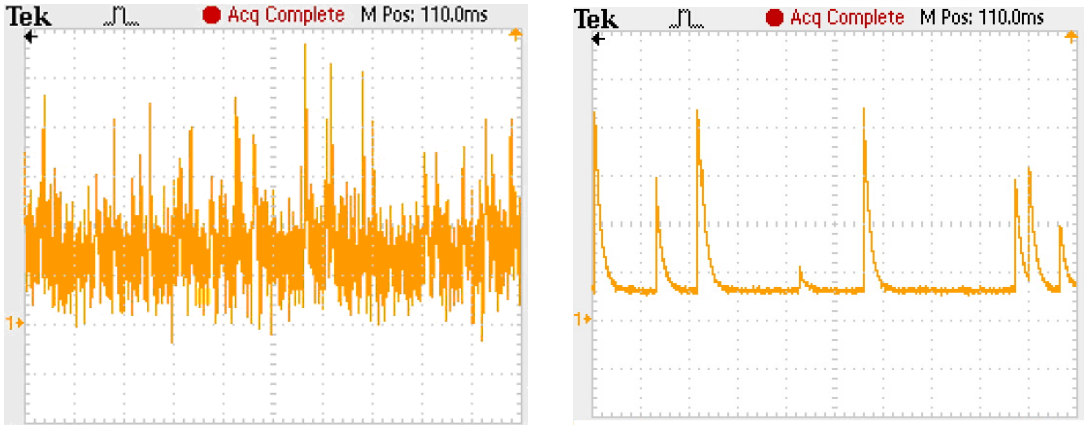


Fig. 3. Measurement of the leakage current of a semiconductor detector.

3.6.3 Energy Resolution, *R*, of the detector.

Conect the detector as shown in Fig.1. Put the triple radioactive source, see pharagraph 3.4.6, inside the vaccum chamber, close to the detector, make vaccum to the measuring chamber, verify the correct polarity and value of the bias voltage according with the data sheet and apply the bias voltage, some pulses will appear in the oscilloscope screen at the preamplifier output, even without the bias voltage. See Fig. 4.a, the pulses could be positive or negative depending of the design of the preamplifier. When the bias voltage is applied, notice that the noise drecreases.



a) b)

Fig. 4.- Pulses at the preamplifier output, a) without the bias voltage; b) with the bias voltage. The settings of the oscilloscope are: 0.5 V/div. and 100 μs/div.

Be sure that the counts be less than 5000 counts per second to avoid errors by pile up or dead time.

Adjust the shaping time of the amplifier to 3 μs or less and adjust the pole-zero for the best unipolar signal at the output of the amplifier, see Fig. 5.

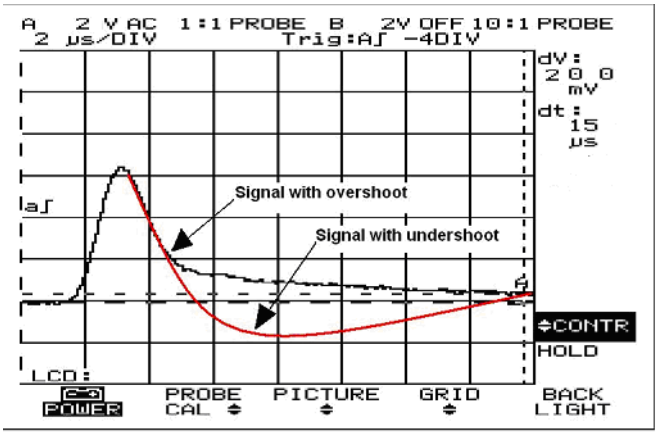


Fig. 5.- Amplifier output signal without proper compensation. The settings of the oscilloscope are: 200 mV/div. and 20 μs/div.

Adjust the amplifier gain to have a pulse of 6 V aproximately, at the output of the amplifier for the Am-241 peak.

Accumulate in the MCA to get more than 10000 counts in the peaks, to minimize the relative uncertainty of the results, < 1%.

Calibrate the MCA in energy units, by using the computer routine for that purpose or obtain the energy calibration equation for the Spectrometry System:

$$y = m\,x + b$$

where: y = energy value m = energy per channel

x = channel value b = energy at channel 0

Employ the three main peaks of the triple source for this porpuse, the energie value of the main peaks are marked in Table 1.

If you made the calibration process using the software routine, then obtain the resolution R, measuring the FWHM in the Am-241 peak, using the MCA.

On the other hand, if you made the calibration manually and you know the value of energy per channel, *m*, get the number of channels at the FWHM and then, convert them to an energy value, see Fig. 6 for the definition of:

$$R = FWHM$$

where:

FWHM = full width at half of maximum in keV

Ho = 5.486 MeV

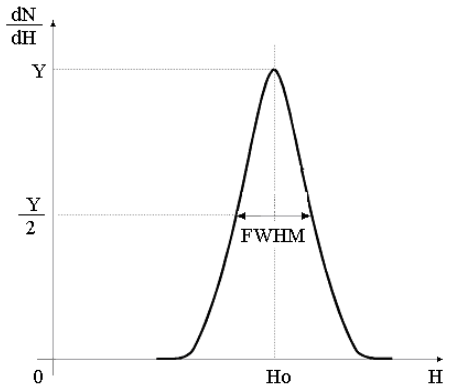


Fig. 6.- Definition of energy resolution.

In the Fig. 7, there is an example of the energy spectrum for a triple source obtained with a charged particle semiconductor detectors in a MCA with 1024 channels. The three peaks of Pu-239,Am-241 and Cm-244 are shadowed.

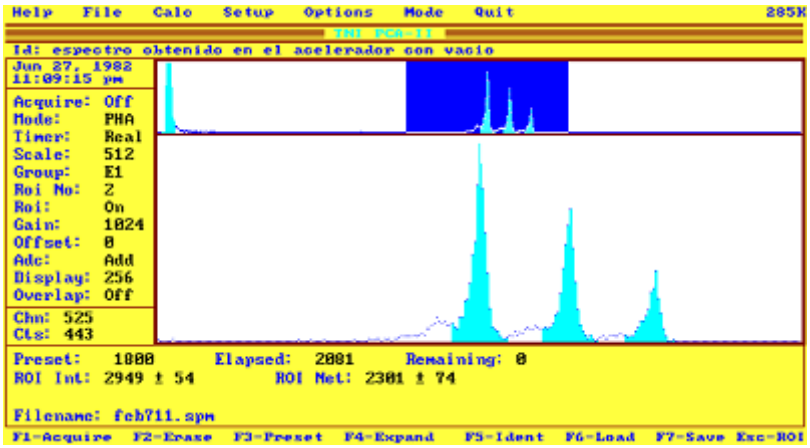


Fig. 7.- Energy spectrum of a triple source obtained with a SSB detector.

In this case, from Fig. 7, the resolution is :

$$R = 23 \text{ keV}$$

3.7.- Detector Aging or damage.

The aging or damage of charged particle semiconductor detectors affects its detection characteristics and parameters. With respect to the functionality, the more important effects are:

- Reduction in the pulse size due to a dirt surface
- Increase of the threshold voltage in forward bias condition and reverse leakage current due to a damage produced by an excessive flux of charged particles.
- Increase of background counts due to radioactive contamination.

4.- ADMINISTRATION OF THE TECHNICAL REPORTS

4.1 Numbering of the reports.

All the generated technical reports must have a unique and consecutive number.

4.2.- Personnel.

The test of charged particle semiconductor detectors must be done by trained personnel.

4.3.- Technical Report of results.

4.3.1 The results of the test of charged particle semiconductor detectors must be registered in a unique technical report, stating the description of the detector, mark, model, serial number, and all the test conditions, including the name of the person who made the tests.

4.3.2 All the technical reports must be classified and keep in a folder for future consult.

5.- ACTION IN CASE OF NON CONFORMITIES.

5.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

5.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

6.- RESPONSIBILITIES

6.1.- Head of the Department.

Supervise that all the activities for testing of charged particle semiconductor detectors follow the established procedure.

6.2.- Area Responsible.

6.2.1 Assure that all the electronic test equipment be in good operational conditions.

6.3.2 Verify that all the activities for testing of charged particle semiconductor detectors follow the established procedure

6.2.3 Verify that the technical reports contain all the details of the testing of charged particle semiconductor detectors.

6.2.4 Maintain a register and control of the technical reports for all the charged particle semiconductor detectors tested in the laboratory.

6.3 Operative Personnel.

6.3.1 Verify that all the electronic test equipment be in good operational conditions.

6.3.2 Follow the steps established in this procedure for the testing of charged particle semiconductor detectors.

6.3.3 Elaborate the technical report of all the tests of the charged particle semiconductor detectors.

6.3.4 Inform to the Area Responsible of any anomalous condition encountered during the test procedure.

7.- BIBLIOGRAPHY

1.- IEEE "Test Procedures for Semiconductor Charged-Particle Detectors". IEEE Std. 300-1988 (R2006), (Revision of IEEE Std 300-1982)

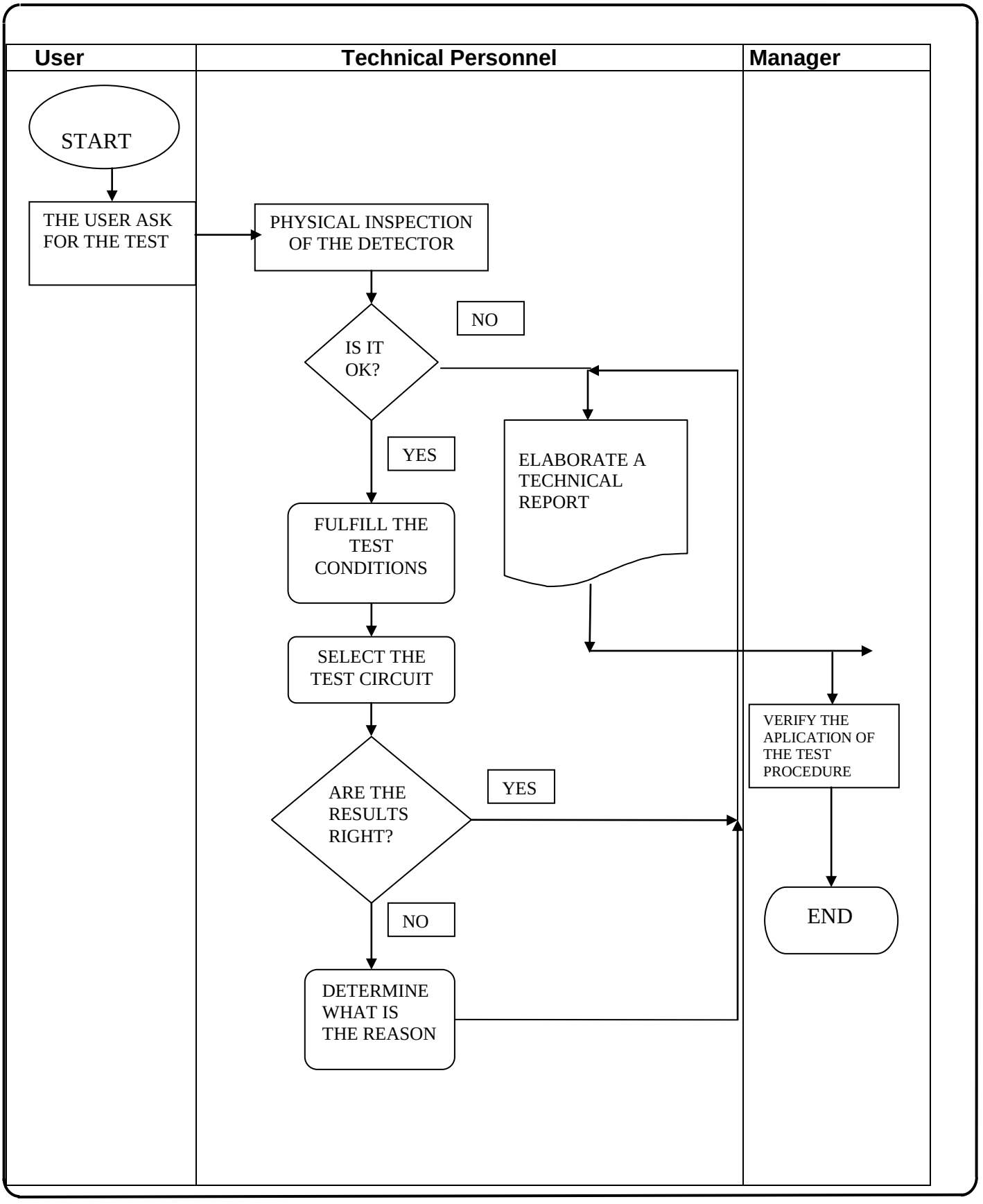
2.- Knoll, Glenn F. "RADIATION DETECTION AND MEASUREMENT", Third Edition, John Wiley and Sons. U.S.A. 2000.

3.- ASTM "Standard Practice for Set-up, Calibration, and Quality Control of Instruments Used for Radioactivity Measurements" ASTM D7282-06 Standard, 2006.

8.- ANNEXES

Annex I

Flow Chart.


User
Technical Personnel
Manager
START
THE USER ASK FOR THE TEST
PHYSICAL INSPECTION OF THE DETECTOR
IS IT OK?
NO
YES
ELABORATE A TECHNICAL REPORT
FULFILL THE TEST CONDITIONS
SELECT THE TEST CIRCUIT
ARE THE RESULTS RIGHT?
YES
NO
DETERMINE WHAT IS THE REASON
VERIFY THE APLICATION OF THE TEST PROCEDURE
END

Annex II

Test report

TEST REPORT Nº CPD-_____

Detector number:_____

Mark:_____ Model:_____
Serial Number:_____Surface:_____

Physical Revision

Condition	YES	NO
Damage in the Body		
Integrity of the surface		
Corrosion		
Oxidation		

Instruments Employed

Instrument	Mark	Model	Serial number
High Voltage Power Supply			
DMM			
Oscilloscope			
Counter, Scaler			
Preamplifier			
Amplifier			
MCA			
Electrometer			
Rate Meter			
Curve tracer			

Environmental Conditions

Temperature	
Pressure (Vacuum)	

Reverse Current

Reverse voltage	
Reverse current	

Radioactive Sources

Source	Serial number	Activity	Date

Test Conditions

Test Voltage (V)	
Polarity of test voltage	
Preamplifier Pulse amplitude (± V)	
Preamplifier Rise time (ns)	
Preamplifier Decay time(μs)	
Time constant of the amplifier (μs)	
Base line restoration	
Amplifier coarse gain	
Amplifier fine gain	
Pile-up rejection	

Draw the signal obtained in the preamplifier output.	***Draw the signal obtained in the amplifier output.***

Results

Resolution (FWHM): _____ measured at 5.486 MeV (peak of Am-241 radioactive source).

Form factor of the pulse (FWTM/FWTM): _____ for 5.486 MeV.

Diagnostic or Comments:

Tested By:	
Date:	